

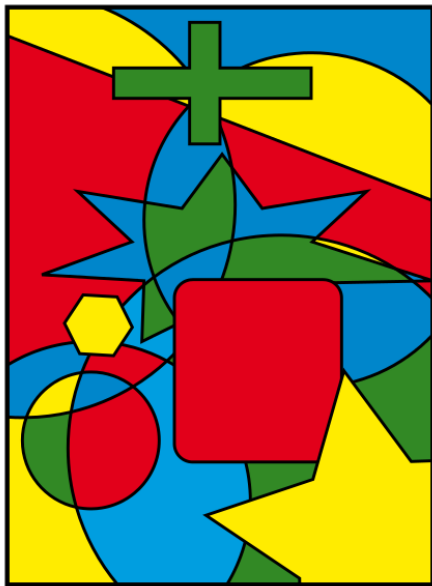
10-planarity-coloring: 平面图与图着色 (Planarity and Coloring)

魏恒峰

hfwei@hnu.edu.cn

2026 年 05 月 21 日



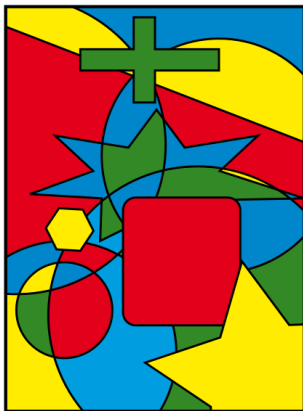


Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent regions** share the same color.*

Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent regions** share the same color.*

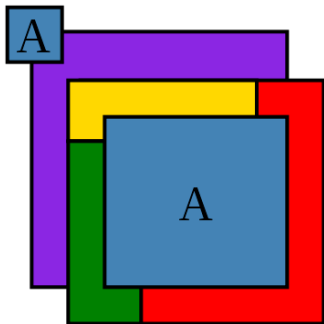


Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent regions** share the same color.*

Theorem (Four Color (Map) Theorem (informal))

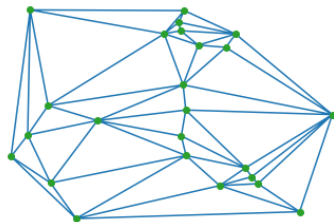
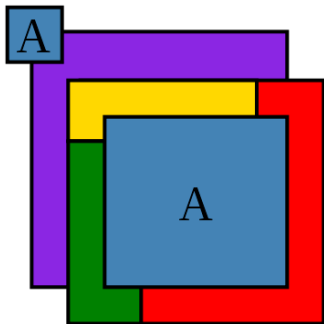
*Every **map** can be colored with only **four** colors such that no two **adjacent** **regions** share the same color.*



Regions should be **contiguous**.

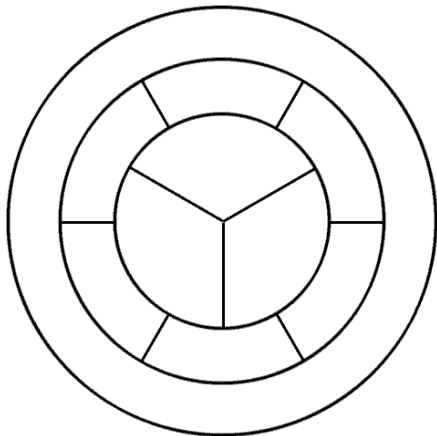
Theorem (Four Color (Map) Theorem (informal))

*Every **map** can be colored with only **four** colors such that no two **adjacent** regions share the same color.*



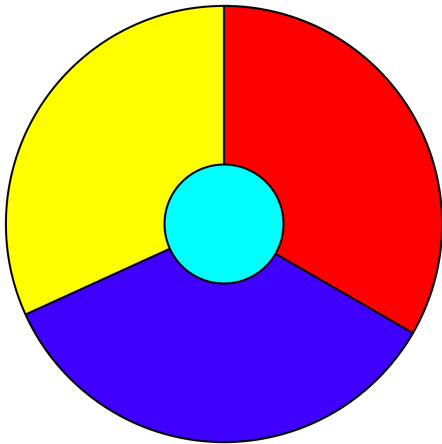
Adjacent regions share a segment.

Regions should be **contiguous**.



4 colors are necessary

4 colors are necessary



Every region is adjacent to the other 3 regions.

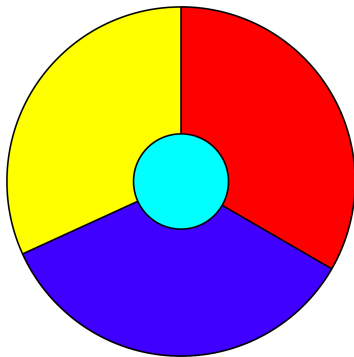
What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

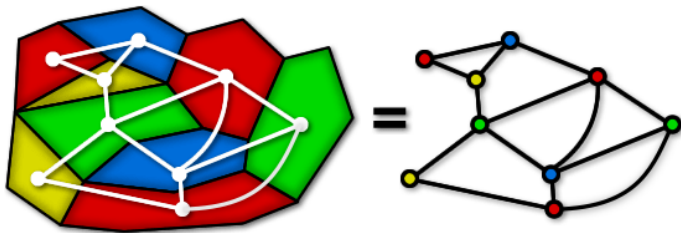
What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

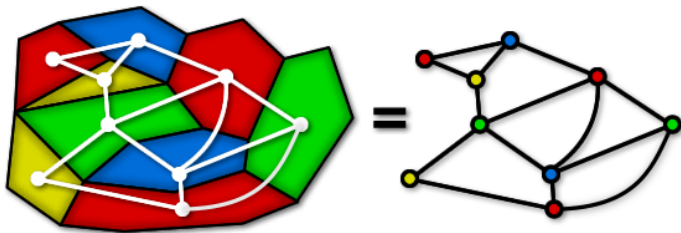


IMPOSSIBLE™

What does Four Color Theorem to do with Graph Theory?



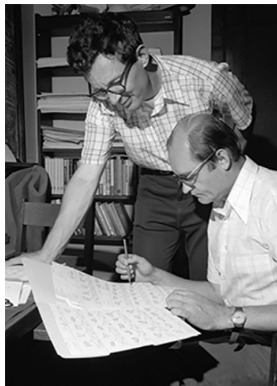




Every map produces a **planar** graph.

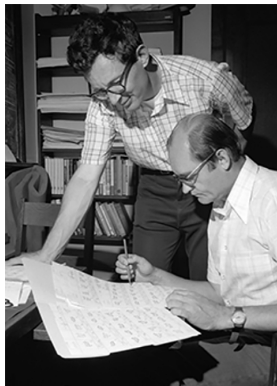
Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every *simple planar* graph is *4-colorable*.



Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every *simple planar* graph is *4-colorable*.



I will *not* show its proof (which I don't understand either)!

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every simple planar graph is 4-colorable.



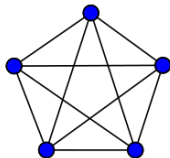
IMPOSSIBLE

What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every *simple planar* graph is *4-colorable*.

IMPOSSIBLE

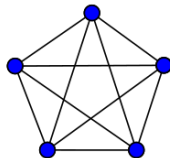


What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every *simple planar* graph is *4-colorable*.

IMPOSSIBLE



What if we have a map which contains 5 regions so that every region is adjacent to the other 4 regions?

Theorem

K_5 is not *planar*.

Theorem

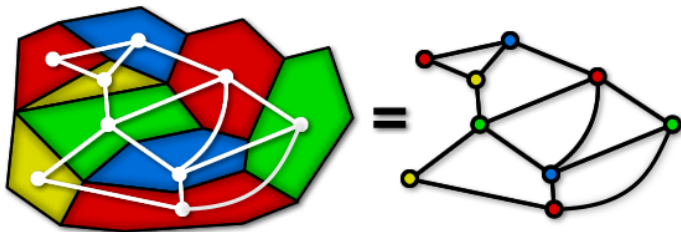
Every simple planar graph is 6-colorable.

Theorem

Every *simple planar* graph is *6-colorable*.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.



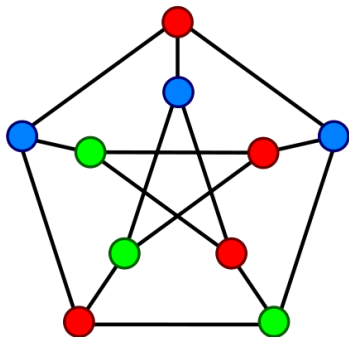
Graph Coloring Problem

Definition (k -Colorable (k -可着色的))

If G is a connected undirected graph without loops, then G is **k -colorable** if its vertices can be colored in k colors so that adjacent vertices have different colors.

Definition (k -Colorable (k -可着色的))

If G is a connected undirected graph without loops, then G is k -colorable if its vertices can be colored in k colors so that adjacent vertices have different colors.



The Petersen graph is ≥ 3 -colorable.

Definition (k -Chromatic (k -色数的))

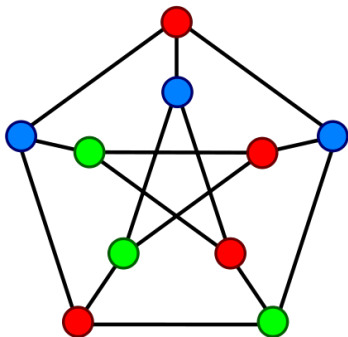
If G is k -colorable, but is not $(k - 1)$ -colorable, then G is k -chromatic.

$$\chi(G) = k$$

Definition (k -Chromatic (k -色数的))

If G is k -colorable, but is not $(k - 1)$ -colorable, then G is k -chromatic.

$$\chi(G) = k$$

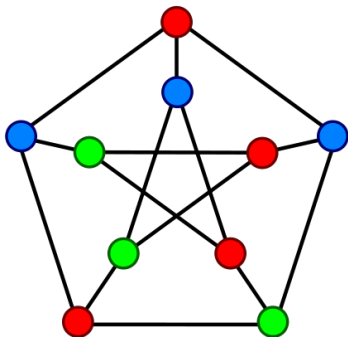


The Petersen graph is 3-chromatic.

Definition (k -Chromatic (k -色数的))

If G is k -colorable, but is not $(k - 1)$ -colorable, then G is k -chromatic.

$$\chi(G) = k$$



The Petersen graph is **3**-chromatic.
(It contains an **odd** cycle.)

Lemma

A graph is 1-colorable iff

Lemma

A graph is 1-colorable iff it is the empty graph with 1 vertex.

Theorem

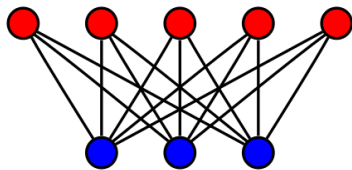
A graph is 2-colorable iff

Theorem

*A graph is 2-colorable iff it is **bipartite**.*

Theorem

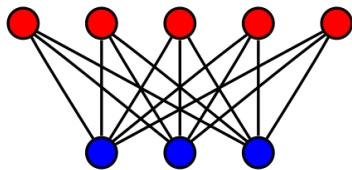
A graph is 2-colorable iff it is *bipartite*.



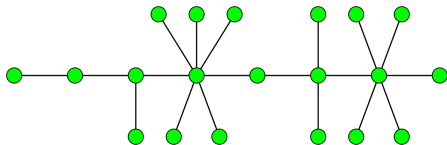
$K_{5,3}$

Theorem

A graph is 2-colorable iff it is *bipartite*.



$K_{5,3}$



Trees are bipartite.

Theorem (Characterization of Bipartite Graphs)

*A graph is 2-colorable iff it is **bipartite**.*

Theorem (Characterization of Bipartite Graphs)

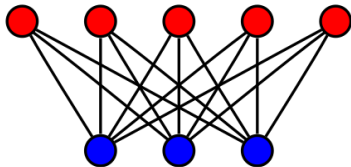
*A graph is 2-colorable iff it is **bipartite**.*

*A graph is **bipartite** iff it does not contain any **odd** cycles.*

Theorem (Characterization of Bipartite Graphs)

A graph is 2-colorable iff it is *bipartite*.

A graph is *bipartite* iff it does not contain any *odd* cycles.

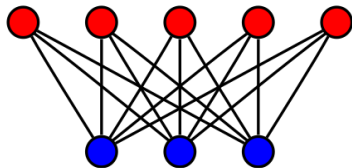


$K_{5,3}$

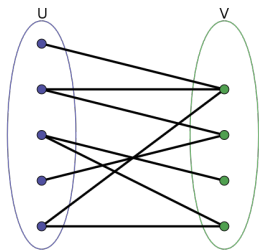
Theorem (Characterization of Bipartite Graphs)

A graph is 2-colorable iff it is *bipartite*.

A graph is *bipartite* iff it does not contain any *odd* cycles.



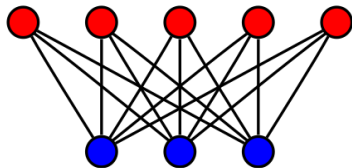
$K_{5,3}$



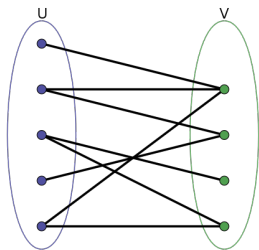
Theorem (Characterization of Bipartite Graphs)

A graph is 2-colorable iff it is *bipartite*.

A graph is *bipartite* iff it does not contain any *odd* cycles.



$K_{5,3}$

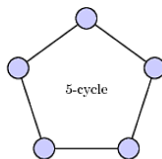
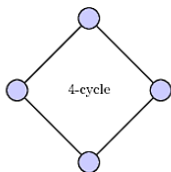
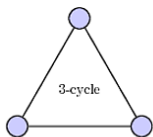


Lemma

Every tree is bipartite and is thus 2-colorable.

Lemma (Characterization of Bipartite Graphs ($\implies$))

If a graph is *bipartite*, then it does not contain any *odd* cycles.



Lemma (Characterization of Bipartite Graphs ($\Leftarrow$))

*If a graph G does not contain any **odd** cycles, then it is **bipartite**.*

Lemma (Characterization of Bipartite Graphs ($\Leftarrow$))

*If a graph G does not contain any **odd** cycles, then it is **bipartite**.*

Choose any vertex $u \in V(G)$.

Lemma (Characterization of Bipartite Graphs ($\Leftarrow$))

If a graph G does not contain any *odd* cycles, then it is *bipartite*.

Choose any vertex $u \in V(G)$.

$X = \{x \in V(G) : \text{the distance from } u \text{ to } x \text{ is even}\}$

$Y = \{y \in V(G) : \text{the distance from } u \text{ to } y \text{ is odd}\}$

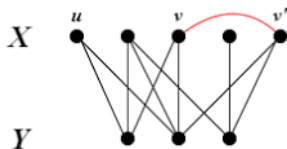
Lemma (Characterization of Bipartite Graphs ($\Leftarrow$))

If a graph G does not contain any *odd* cycles, then it is *bipartite*.

Choose any vertex $u \in V(G)$.

$X = \{x \in V(G) : \text{the distance from } u \text{ to } x \text{ is even}\}$

$Y = \{y \in V(G) : \text{the distance from } u \text{ to } y \text{ is odd}\}$



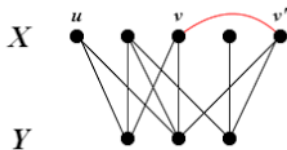
Lemma (Characterization of Bipartite Graphs ($\Leftarrow$))

If a graph G does not contain any **odd** cycles, then it is **bipartite**.

Choose any vertex $u \in V(G)$.

$X = \{x \in V(G) : \text{the distance from } u \text{ to } x \text{ is even}\}$

$Y = \{y \in V(G) : \text{the distance from } u \text{ to } y \text{ is odd}\}$



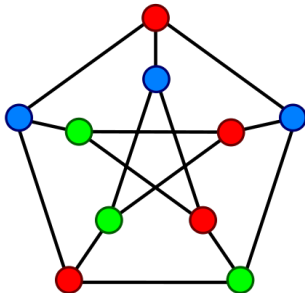
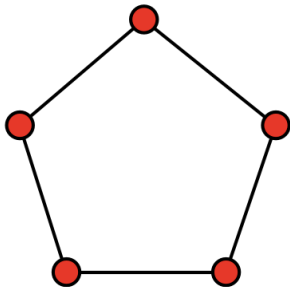
$u \rightsquigarrow v \rightarrow v' \rightsquigarrow u$ is an **odd** cycle.

Theorem

The 3-coloring problem (i.e., testing whether a graph is 3-colorable or not) is NP-complete (HARD!).

Theorem

The 3-coloring problem (i.e., testing whether a graph is 3-colorable or not) is NP-complete (HARD!).



Theorem

The 4-coloring problem is NP-complete (HARD!).

Theorem

The 4-coloring problem is NP-complete (HARD!).

Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every simple planar graph is 4-colorable.

Theorem

The 4-coloring problem is NP-complete (HARD!).



Theorem (Four Color Theorem (Kenneth Appel, Wolfgang Haken; 1976))

Every simple planar graph is 4-colorable.

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

By induction on the number of vertices of G .

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

By induction on the number of vertices of G .

Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

By induction on the number of vertices of G .

Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Induction Hypothesis: Suppose that for any simple connected graph G with n vertices,

$$\chi(G) \leq \Delta(G) + 1.$$

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

By induction on the number of vertices of G .

Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Induction Hypothesis: Suppose that for any simple connected graph G with n vertices,

$$\chi(G) \leq \Delta(G) + 1.$$

Induction Step: Consider a simple connected graph G with $n + 1$ vertices.

Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

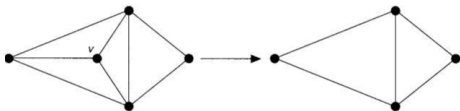
By induction on the number of vertices of G .

Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Induction Hypothesis: Suppose that for any simple connected graph G with n vertices,

$$\chi(G) \leq \Delta(G) + 1.$$

Induction Step: Consider a simple connected graph G with $n + 1$ vertices.



Theorem

Let G be a simple connected graph. Then,

$$\chi(G) \leq \Delta(G) + 1.$$

By induction on the number of vertices of G .

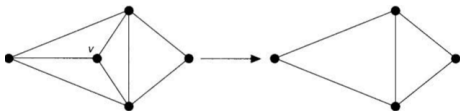
Basis Step: $n = 1$. $\chi(G) = 1$ and $\Delta(G) = 0$.

Induction Hypothesis: Suppose that for any simple connected graph G with n vertices,

$$\chi(G) \leq \Delta(G) + 1.$$

Induction Step: Consider a simple connected graph G with $n + 1$ vertices.

$$\deg(v) \leq \Delta(G)$$



Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

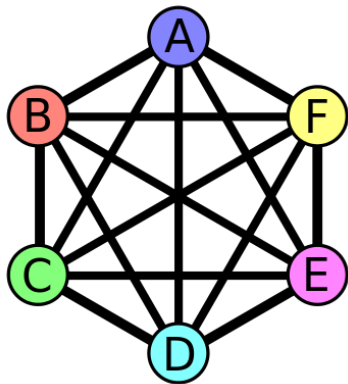
Let G be a *simple* connected graph other than a *complete graph* or an *odd cycle*. Then

$$\chi(G) \leq \Delta(G).$$

Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

Let G be a **simple** connected graph other than a **complete graph** or an **odd cycle**. Then

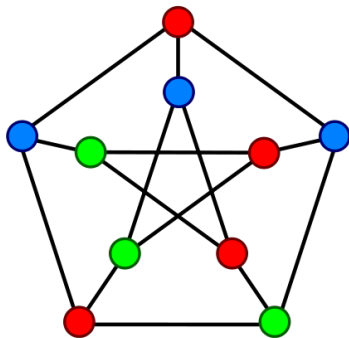
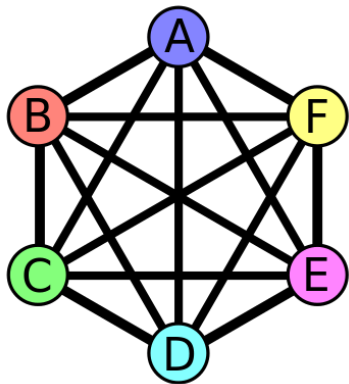
$$\chi(G) \leq \Delta(G).$$



Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

Let G be a *simple* connected graph other than a *complete graph* or an *odd cycle*. Then

$$\chi(G) \leq \Delta(G).$$



Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

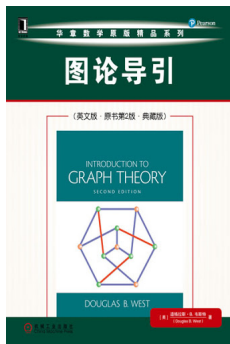
Let G be a *simple* connected graph other than a *complete graph* or an *odd cycle*. Then

$$\chi(G) \leq \Delta(G).$$

Theorem (Brooks's Theorem (R. Leonard Brooks; 1941))

Let G be a *simple* connected graph other than a *complete graph* or an *odd cycle*. Then

$$\chi(G) \leq \Delta(G).$$



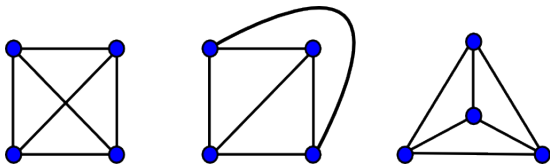
Theorem 5.1.22

Definition (Planar Graph (平面图))

A **planar graph** is a graph that **can** be drawn in the plane without **edge crossings**.

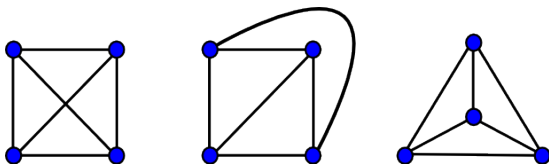
Definition (Planar Graph (平面图))

A **planar graph** is a graph that **can** be drawn in the plane without **edge crossings**.



Definition (Planar Graph (平面图))

A **planar graph** is a graph that **can** be drawn in the plane without **edge crossings**.

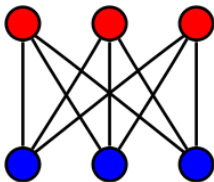


Theorem (K. Wagner (1936); I. Fáry (1948))

*Every simple planar graph can be drawn with **straight lines**.*

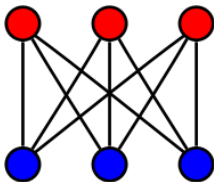
Theorem (Kazimierz Kuratowski, 1930)

The *utility graph* $K_{3,3}$ is non-planar.



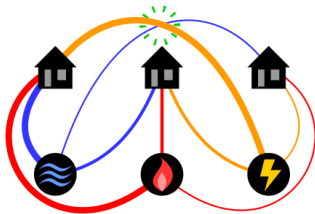
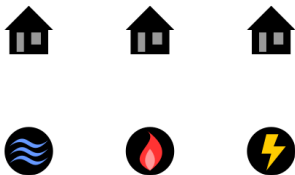
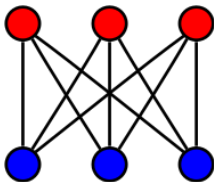
Theorem (Kazimierz Kuratowski, 1930)

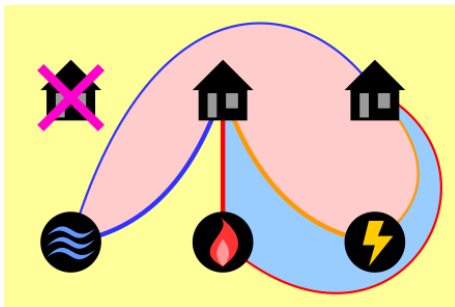
The *utility graph* $K_{3,3}$ is non-planar.



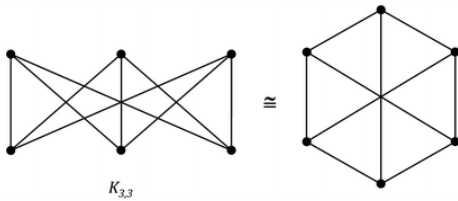
Theorem (Kazimierz Kuratowski, 1930)

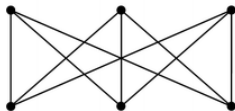
The *utility graph* $K_{3,3}$ is non-planar.





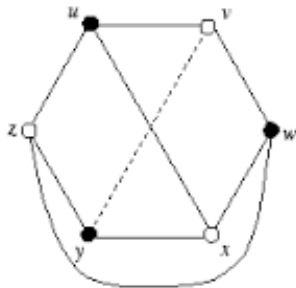
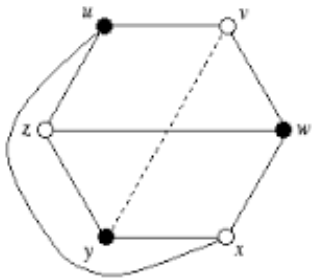
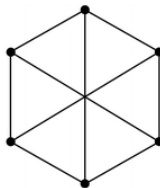
Proof without Words

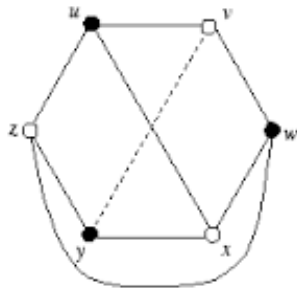
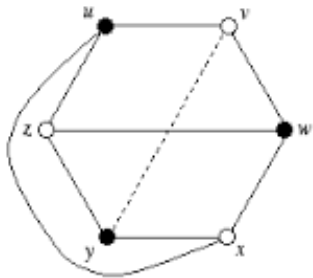
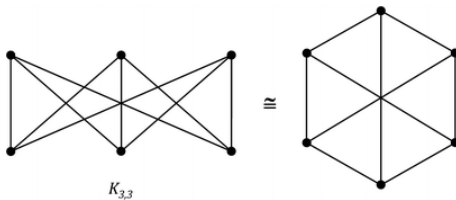




$K_{3,3}$

$\cong$

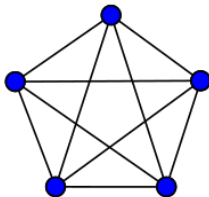




$$\text{cr}(K_{3,3}) = 1 \quad (\text{crossing number})$$

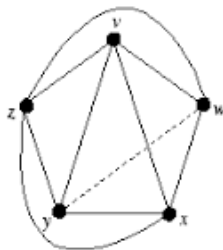
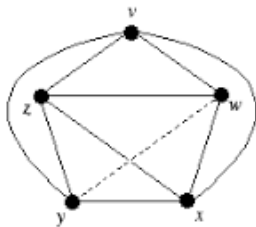
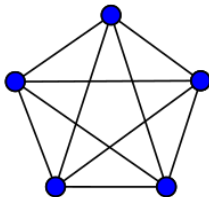
Theorem

K_5 is non-planar.



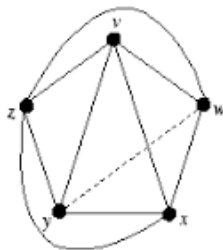
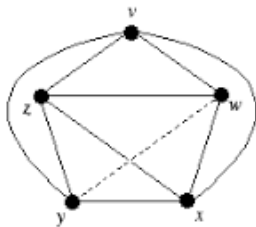
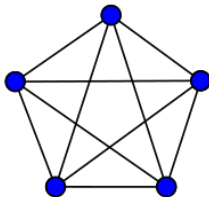
Theorem

K_5 is non-planar.



Theorem

K_5 is non-planar.



$$\text{cr}(K_5) = 1$$

Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.

Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.



Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no *subgraph homeomorphic to K_5 or $K_{3,3}$* .



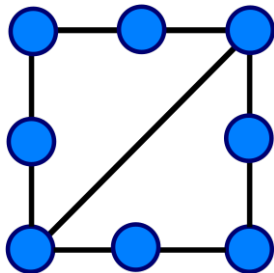
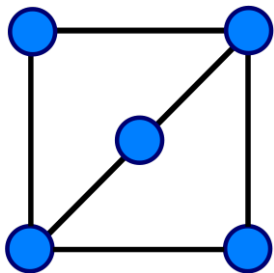
*“The K in K_5 stands for Kazimierz,
and the K in $K_{3,3}$ stands for Kuratowski.”*

Definition (Homeomorphic)

Two graphs are **homeomorphic** if one can be obtained from another by **inserting or contracting** vertices of **degree 2**.

Definition (Homeomorphic)

Two graphs are **homeomorphic** if one can be obtained from another by **inserting or contracting** vertices of **degree 2**.

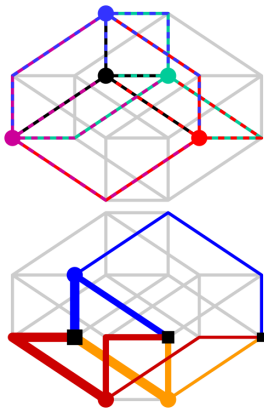


Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.

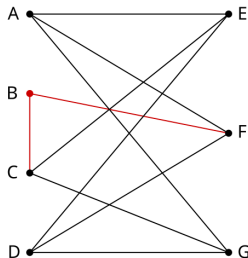
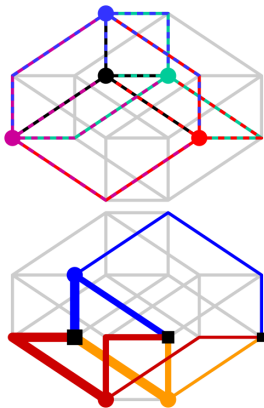
Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no subgraph homeomorphic to K_5 or $K_{3,3}$.



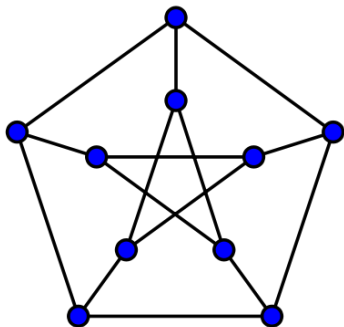
Theorem (Kazimierz Kuratowski, 1930)

A graph is planar iff it contains no *subgraph homeomorphic to K_5 or $K_{3,3}$* .



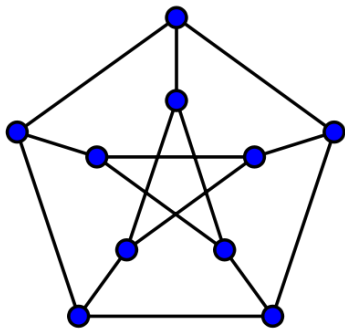
Theorem

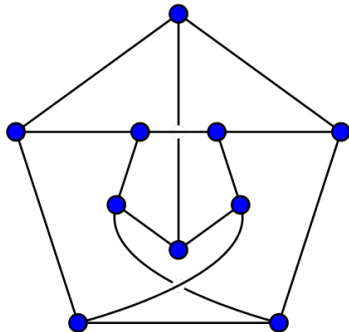
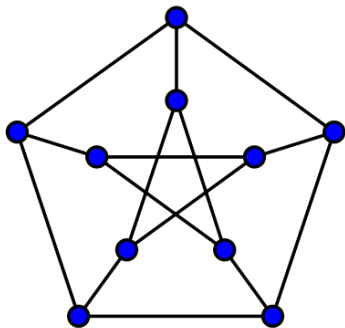
The Petersen graph is non-planar.

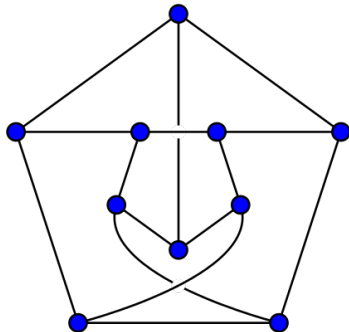
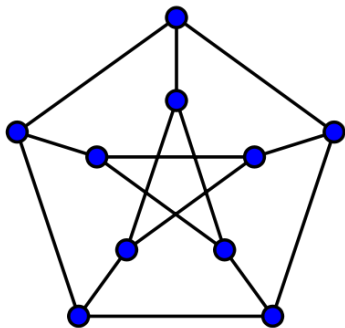


[https:](https://github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif)

[//github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif](https://github.com/courses-at-hnu-by-hfwei/math4cs-lectures/blob/main/10-planarity-coloring/figs/Kuratowski-Petersen.gif)







$$\text{cr}(\text{Petersen Graph}) = 2$$

Efficient Planarity Testing

JOHN HOPCROFT AND ROBERT TARJAN

Cornell University, Ithaca, New York

ABSTRACT. This paper describes an efficient algorithm to determine whether an arbitrary graph G can be embedded in the plane. The algorithm may be viewed as an iterative version of a method originally proposed by Auslander and Parter and correctly formulated by Goldstein. The algorithm uses depth-first search and has $O(V)$ time and space bounds, where V is the number of vertices in G . An ALGOL implementation of the algorithm successfully tested graphs with as many as 900 vertices in less than 12 seconds.

KEY WORDS AND PHRASES: algorithm, complexity, depth-first search, embedding, genus, graph, palm tree, planarity, spanning tree

CR CATEGORIES: 3.24, 5.25, 5.32



A planar graph should not has too many edges.

Theorem (Euler's Formula, 1750)

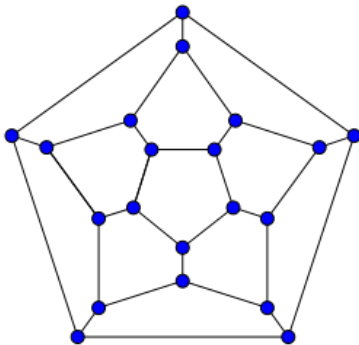
Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$

Theorem (Euler's Formula, 1750)

Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

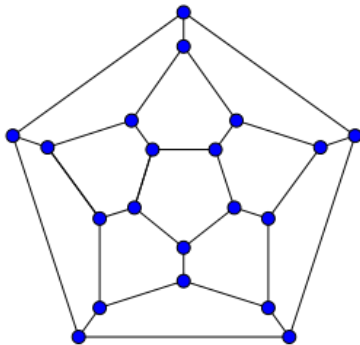
$$n - m + f = 2$$



Theorem (Euler's Formula, 1750)

Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$



$$n - m + f = 20 - 30 + 12 = 2$$

Theorem (Euler's Formula, 1750)

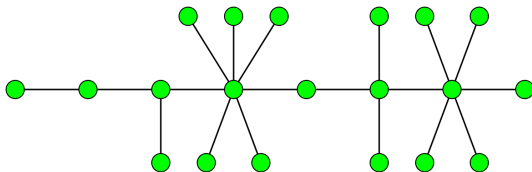
Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$

Theorem (Euler's Formula, 1750)

Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

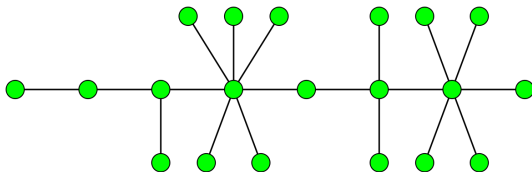
$$n - m + f = 2$$



Theorem (Euler's Formula, 1750)

Let G be a *plane drawing* of a *connected* planar graph, and let n , m , and f denote respectively the number of vertices, edges, and *faces* of G .

$$n - m + f = 2$$



$$n - m + f = n - (n - 1) + 1 = 2$$

By induction on the number of edges of G .

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.
If G is a tree, we are done.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

Let e be an edge in some cycle of G .

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

Let e be an edge in some cycle of G .

Consider $G' = G - e$.

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

Let e be an edge in some cycle of G .

Consider $G' = G - e$.

$$n - (m - 1) + (f - 1) = 2$$

By induction on the number of edges of G .

Basis Step: $m = 0$. We have $n = 1$ and $f = 1$.

Induction Hypothesis: It holds for plane graphs with m edges.

Induction Step: Consider a plane graph G with $m + 1$ edges.

If G is a tree, we are done.

Otherwise, G contains a cycle.

Let e be an edge in some cycle of G .

Consider $G' = G - e$.

$$n - (m - 1) + (f - 1) = 2$$

Therefore,

$$n - m + f = 2$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

$$n - m + f = 2$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

$$n - m + f = 2$$

$$2m = \sum_f \deg(f) \geq 3f$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. Then

$$m \leq 3n - 6.$$

$$n - m + f = 2$$

$$2m = \sum_f \deg(f) \geq 3f$$

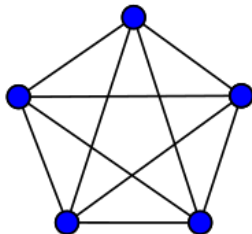
Double Counting the Edges around Faces:

each edge bounds 2 faces;

each face is bounded by ≥ 3 edges

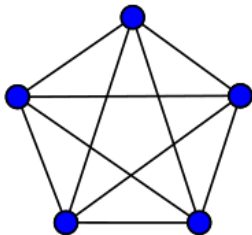
Theorem

K_5 is non-planar.



Theorem

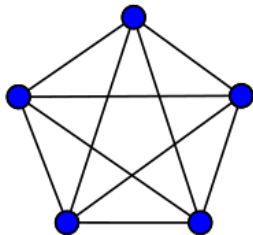
K_5 is non-planar.



$$m \leq 3n - 6$$

Theorem

K_5 is non-planar.



$$m \leq 3n - 6$$

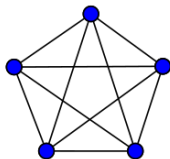
$$10 \leq 3 \times 5 - 6$$

Theorem

K_5 is non-planar.

Theorem

K_5 is non-planar.

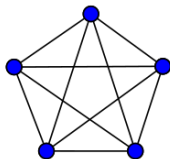


$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$

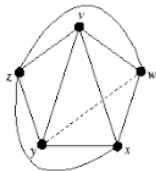
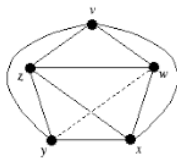
Theorem

K_5 is non-planar.



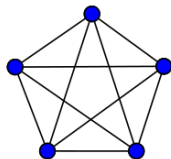
$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$



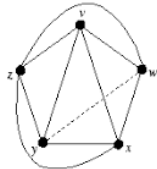
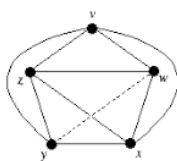
Theorem

K_5 is non-planar.

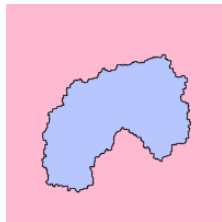


$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$

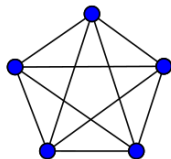


Theorem (Jordan Curve Theorem (1887))



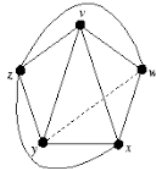
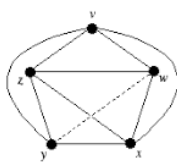
Theorem

K_5 is non-planar.

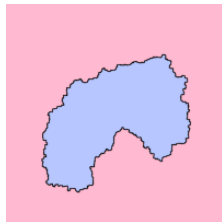


$$m \leq 3n - 6$$

$$10 \leq 3 \times 5 - 6$$

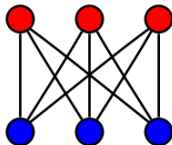


Theorem (Jordan Curve Theorem (1887))



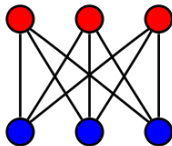
Theorem

$K_{3,3}$ is non-planar.



Theorem

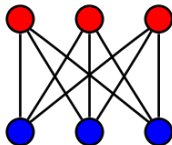
$K_{3,3}$ is non-planar.



$$m \leq 3n - 6$$

Theorem

$K_{3,3}$ is non-planar.

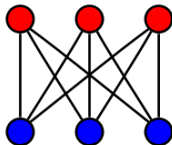


$$m \leq 3n - 6$$

$$9 \leq 3 \times 6 - 6$$

Theorem

$K_{3,3}$ is non-planar.



$$m \leq 3n - 6$$

$$9 \leq 3 \times 6 - 6$$

FAILED

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. *If G has no triangles, then*

$$m \leq 2n - 4.$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. *If G has no triangles, then*

$$m \leq 2n - 4.$$

$$n - m + f = 2$$

Theorem

Let G be a simple connected planar graph with $n \geq 3$ vertices and m edges. *If G has no triangles, then*

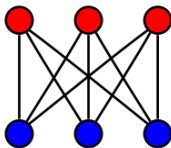
$$m \leq 2n - 4.$$

$$n - m + f = 2$$

$$2m = \sum_f \deg(f) \geq 4f$$

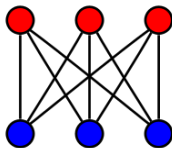
Theorem

$K_{3,3}$ is non-planar.



Theorem

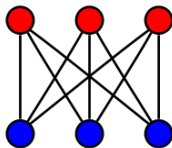
$K_{3,3}$ is non-planar.



$$m \leq 2n - 4$$

Theorem

$K_{3,3}$ is non-planar.



$$m \leq 2n - 4$$

$$9 \leq 2 \times 6 - 4$$

Theorem

Every simple planar graph contains a vertex of degree ≤ 5 .

Theorem

Every simple planar graph contains a vertex of degree ≤ 5 .

Suppose that, **by contradiction**, $\delta(G) \geq 6$.

Theorem

Every simple planar graph contains a vertex of degree ≤ 5 .

Suppose that, **by contradiction**, $\delta(G) \geq 6$.

$$2m = \sum_v \deg(v) \geq 6n$$

Theorem

Every simple planar graph contains a vertex of degree ≤ 5 .

Suppose that, **by contradiction**, $\delta(G) \geq 6$.

$$2m = \sum_v \deg(v) \geq 6n$$

$$m \leq 3n - 6$$

Theorem

Every simple planar graph contains a vertex of degree ≤ 5 .

Suppose that, **by contradiction**, $\delta(G) \geq 6$.

$$2m = \sum_v \deg(v) \geq 6n$$

$$m \leq 3n - 6$$

$$3n \leq m \leq 3n - 6$$

Theorem

Every simple planar graph is 6-colorable.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is 6-colorable.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is 6-colorable.

Thus, G is 6-colorable.

Theorem

Every *simple planar* graph is *6-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is 6-colorable.

Thus, G is 6-colorable. ($\deg(v) \leq 5$)

Theorem (Percy John Heawood (1890))

Every simple planar graph is 5-colorable.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is *5-colorable*.

Theorem (Percy John Heawood (1890))

Every *simple planar* graph is *5-colorable*.

By induction on the number of vertices.

Basis Step: $n = 1$. Trivial.

Induction Hypothesis: Suppose that it holds for simple planar graphs with $n \geq 1$ vertices.

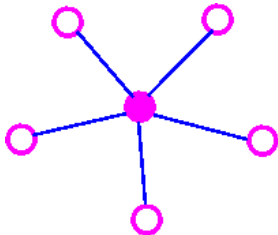
Induction Step: Consider a simple planar graph G with $n + 1$ vertices.

G contains a vertex v of degree ≤ 5 .

$G' = G - v$ is *5-colorable*.

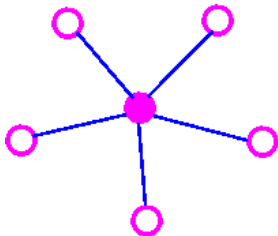
If $\deg(v) < 5$, G is 5-colorable.

Now assume that $\deg(v) = 5$.



$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

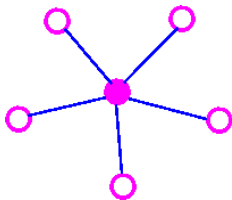
Now assume that $\deg(v) = 5$.



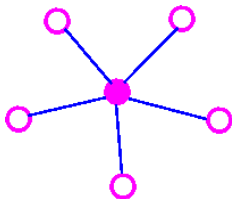
$$\{v, v_1\}, \{v, v_2\}, \{v, v_3\}, \{v, v_4\}, \{v, v_5\}$$

If v_1, v_2, v_3, v_4 , and v_5 uses < 5 colors, we are done.

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.

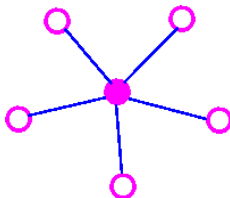


Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

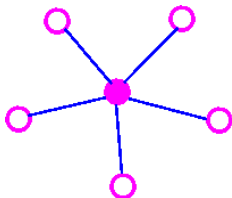
Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a **red-blue** path.

Now assume that v_1 , v_2 , v_3 , v_4 , and v_5 uses 5 colors.

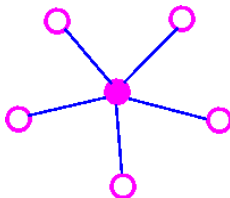


Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a **red-blue** path.

$$v_1 \in S, \quad v_3 \notin S$$

Now assume that v_1, v_2, v_3, v_4 , and v_5 uses 5 colors.



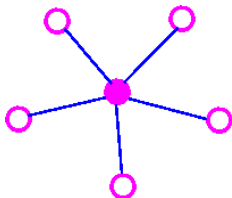
Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a **red-blue** path.

$$v_1 \in S, \quad v_3 \notin S$$

Interchange the colors of the **red** or **blue** vertices in S

Now assume that v_1, v_2, v_3, v_4 , and v_5 uses 5 colors.



Suppose that there is *no* $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored **red** or **blue**.

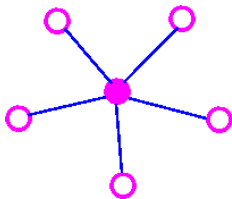
Let S be the set of all **red** or **blue** vertices of $G - v$
connected to v_1 by a **red-blue** path.

$$v_1 \in S, \quad v_3 \notin S$$

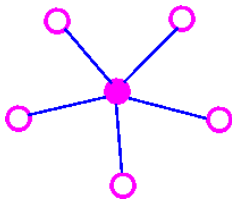
Interchange the colors of the **red** or **blue** vertices in S

Coloring v **red** produces a 5-coloring of G .

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.

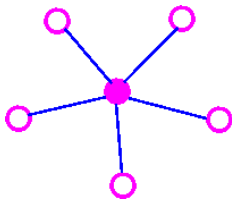


Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.



There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

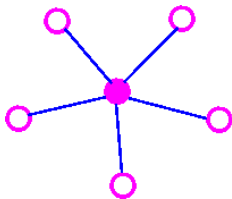
Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.



There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

(Otherwise, G' and thus G is non-planar.)

Now assume that there *is* a $v_1 \sim v_3$ path in $G' = G - v$,
all of whose vertices are colored red or blue.



There cannot be $v_2 \sim v_4$ path in $G' = G - v$,
all of whose vertices are colored green or purple.

(Otherwise, G' and thus G is non-planar.)

By similar argument, G is 5-colorable.

$P(G, k)$: How many ways to color the vertices of G with k colors?

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) =$$

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$$P(C_4, k) =$$

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$$P(C_4, k) = k(k-1)^2(k-2)$$

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$$P(C_4, k) = k(k-1)^2(k-2)$$

$$P(K_4, k) =$$

$P(G, k)$: How many ways to color the vertices of G with k colors?

$$P(P_4, k) = k(k-1)^3$$

$$P(C_4, k) = k(k-1)^2(k-2)$$

$$P(K_4, k) = k(k-1)(k-2)(k-3)$$

Theorem (Deletion-Contraction Recurrence)

Let G be a simple graph and e be an edge of G . Then

$$P(G, k) = P(G - e, k) - P(G/e, k),$$

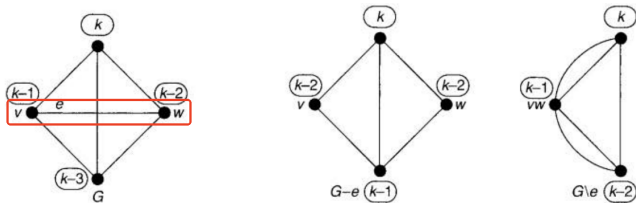
where $G - e$ is the graph obtained from G by *deleting* e ,
and G/e is the graph obtained from G by *contracting* e .

Theorem (Deletion-Contraction Recurrence)

Let G be a simple graph and e be an edge of G . Then

$$P(G, k) = P(G - e, k) - P(G/e, k),$$

where $G - e$ is the graph obtained from G by *deleting* e ,
and G/e is the graph obtained from G by *contracting* e .

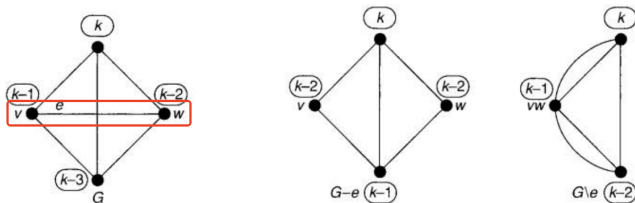


Theorem (Deletion-Contraction Recurrence)

Let G be a simple graph and e be an edge of G . Then

$$P(G, k) = P(G - e, k) - P(G/e, k),$$

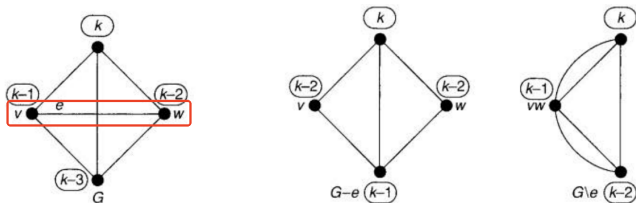
where $G - e$ is the graph obtained from G by *deleting* e ,
and G/e is the graph obtained from G by *contracting* e .



$$k(k-1)(k-2)(k-3) = k(k-1)(k-2)^2 - k(k-1)(k-2)$$

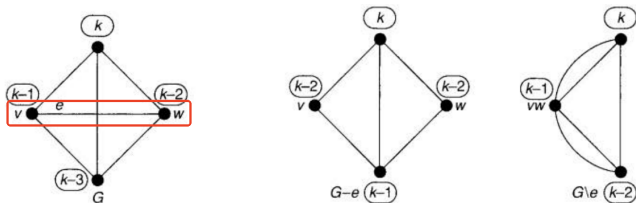
Theorem (Deletion-Contraction Recurrence)

$$P(G, k) = P(G - e, k) - P(G/e, k)$$



Theorem (Deletion-Contraction Recurrence)

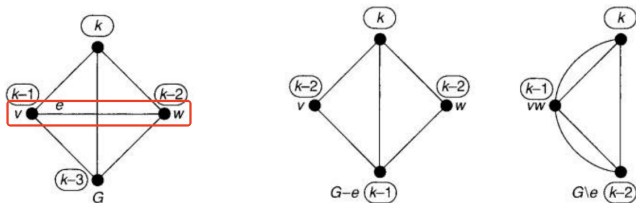
$$P(G, k) = P(G - e, k) - P(G/e, k)$$



$$P(G - e, k) = P(G, k) + P(G/e, k)$$

Theorem (Deletion-Contraction Recurrence)

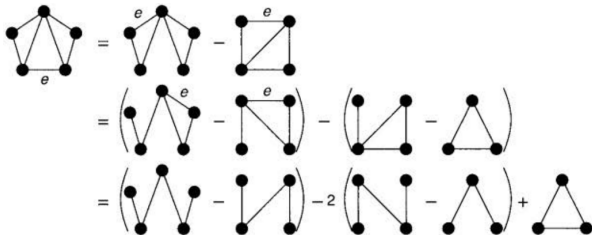
$$P(G, k) = P(G - e, k) - P(G/e, k)$$



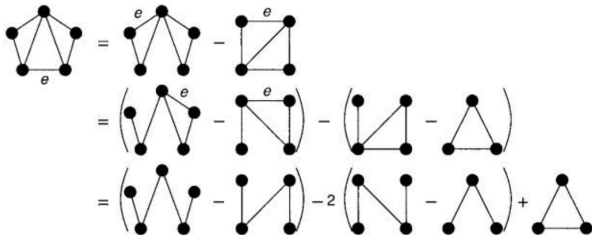
$$P(G - e, k) = P(G, k) + P(G/e, k)$$

$P(G - e, k) = P(G - e, k)$ with v and w having *different* colors
 + $P(G - e, k)$ with v and w having the *same* color

$$P(G, k) = P(G - e, k) - P(G/e, k)$$

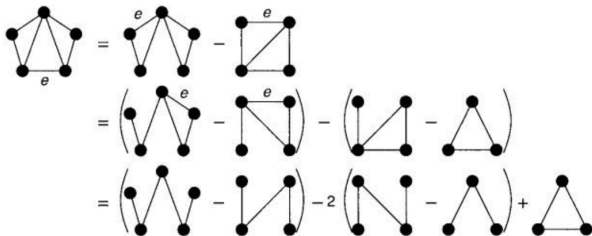


$$P(G, k) = P(G - e, k) - P(G/e, k)$$



$$\begin{aligned} P(G, k) &= k(k-1)^4 - 3k(k-1)^3 + 2k(k-1)^2 + k(k-1)(k-2) \\ &= k^5 - 7k^4 + 18k^3 - 20k^2 + 8k \end{aligned}$$

$$P(G, k) = P(G - e, k) - P(G/e, k)$$

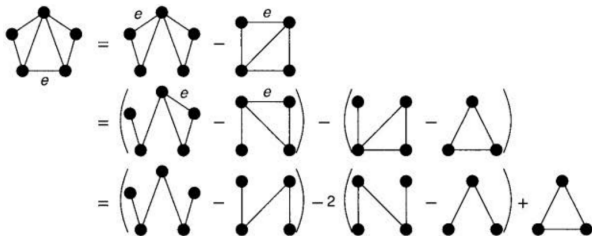


$$\begin{aligned} P(G, k) &= k(k-1)^4 - 3k(k-1)^3 + 2k(k-1)^2 + k(k-1)(k-2) \\ &= k^5 - 7k^4 + 18k^3 - 20k^2 + 8k \end{aligned}$$

Theorem (Chromatic Polynomial)

$P(G, k)$ is a polynomial in k .

$$P(G, k) = P(G - e, k) - P(G/e, k)$$



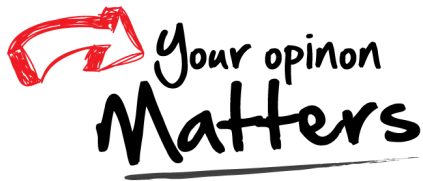
$$\begin{aligned} P(G, k) &= k(k-1)^4 - 3k(k-1)^3 + 2k(k-1)^2 + k(k-1)(k-2) \\ &= k^5 - 7k^4 + 18k^3 - 20k^2 + 8k \end{aligned}$$

Theorem (Chromatic Polynomial)

$P(G, k)$ is a polynomial in k .

$$k^n - mk^{n-1} + \dots - \dots + ak + 0$$

Thank
You!



hfwei@hnu.edu.cn